

# Stock Market Database Schema Documentation

This document contains the SQLite database schema for the stock market analytics and AI equity research platform. It includes core financial tables, market data structures, ownership tracking, technical indicators, macro indicators, growth metrics, balance sheet data, cash flow structures, and indexing strategy.

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-- SQLite schema for stock market database
-- Generated from SQLite .schema output

-- Core stocks table
CREATE TABLE stocks (
  symbol      TEXT PRIMARY KEY,
  name        TEXT,
  exchange    TEXT DEFAULT 'NSE',
  sector      TEXT,
  industry    TEXT,
  currency    TEXT DEFAULT 'INR',
  created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Daily price data
CREATE TABLE price_daily (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  symbol      TEXT NOT NULL REFERENCES stocks(symbol),
  date        DATE NOT NULL,
  open        REAL,
  high        REAL,
  low         REAL,
  close       REAL,
  adj_close   REAL,
  volume      INTEGER,
  UNIQUE (symbol, date)
);

-- Intraday price data
CREATE TABLE price_intraday (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  symbol      TEXT NOT NULL REFERENCES stocks(symbol),
  ts          TIMESTAMP NOT NULL,
  interval    TEXT NOT NULL DEFAULT '1m',
  open        REAL, high REAL, low REAL, close REAL, volume INTEGER,
  UNIQUE (symbol, ts, interval)
);

-- Fundamentals data
CREATE TABLE fundamentals (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  symbol      TEXT NOT NULL REFERENCES stocks(symbol),
  as_of_date  DATE NOT NULL,

  -- Screener overview (most authoritative – scraped directly)
  current_price REAL,
  face_value    REAL,
  high_52w      REAL,
  low_52w       REAL,
  book_value    REAL,

  -- Profitability
  roe_pct       REAL,
  roce_pct      REAL,
  roa_pct       REAL,
  interest_coverage REAL,

  -- Cash flow (Rs. Crores)
  gross_margin_pct REAL,
  net_profit_margin_pct REAL,
  ebitda_margin_pct REAL,
  ebit_margin_pct  REAL,
  opm_pct         REAL,

  -- Leverage & liquidity
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    debt_to_equity            REAL,
    current_ratio             REAL,
    quick_ratio               REAL,

    -- Working capital efficiency
    dso_days                  REAL,
    dio_days                  REAL,
    dpo_days                  REAL,
    cash_conversion_cycle     REAL,
    working_capital_days      REAL,

    -- Valuation
    eps_annual                REAL,
    pe_ratio                  REAL,
    pb_ratio                  REAL,
    graham_number             REAL,
    dividend_yield_pct        REAL,
    dividend_payout_pct       REAL,
    forward_pe                REAL,

    -- Scale (Rs. Crores)
    market_cap                REAL,
    revenue                   REAL,
    ebitda                    REAL,
    inventory                  REAL,
    ev                        REAL,
    ttm_eps                   REAL,
    ttm_pe                    REAL,
    ev_ebitda                 REAL,
    ev_revenue                 REAL,

    -- Growth
    earnings_growth_json      TEXT,

    -- Data quality
    data_source                TEXT DEFAULT 'yfinance',
    completeness_pct           REAL,

    UNIQUE (symbol, as_of_date)
);

-- Quarterly results
CREATE TABLE quarterly_results (
    id                        INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol                    TEXT NOT NULL REFERENCES stocks(symbol),
    period_end                DATE NOT NULL,

    -- P&L (Rs. Crores)
    sales                     REAL NOT NULL,
    expenses                  REAL,
    operating_profit          REAL,
    opm_pct                   REAL,
    other_income              REAL,
    interest                  REAL,
    depreciation              REAL,
    profit_before_tax         REAL,
    tax_pct                   REAL,
    net_profit                REAL NOT NULL,
    eps                       REAL,

    -- Data quality
    data_source                TEXT DEFAULT 'Screener.in',
    is_audited                 INTEGER DEFAULT 0,
    completeness_pct           REAL,

    UNIQUE (symbol, period_end)
);

-- Annual results
CREATE TABLE annual_results (
    id                        INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol                    TEXT NOT NULL REFERENCES stocks(symbol),
    period_end                DATE NOT NULL,

    -- P&L (Rs. Crores)
    sales                     REAL NOT NULL,
    expenses                  REAL,
    operating_profit          REAL,
    opm_pct                   REAL,

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        other_income            REAL,
        interest                REAL,
        depreciation            REAL,
        profit_before_tax       REAL,
        tax_pct                 REAL,
        net_profit              REAL NOT NULL,
        eps                     REAL,
        dividend_payout_pct     REAL,

        -- Data quality
        source                   TEXT DEFAULT 'Screener.in',
        completeness_pct        REAL,
        is_ttm                   INTEGER DEFAULT 0,
        data_source              TEXT DEFAULT 'screener',

        UNIQUE (symbol, period_end)
    );

-- Annual cashflow derived
CREATE TABLE "annual_cashflow_derived" (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol             TEXT NOT NULL REFERENCES stocks(symbol),
    annual_end         DATE NOT NULL,

    revenue            REAL,
    net_income          REAL,
    dna                 REAL,
    approx_op_cf        REAL,
    approx_capex        REAL,
    approx_fcf          REAL,
    fcf_margin_pct      REAL,

    capex_source        TEXT,
    quality_score        INTEGER DEFAULT 1,
    is_real              INTEGER DEFAULT 0,
    is_interpolated      INTEGER DEFAULT 0,
    data_note            TEXT,
    unit                 TEXT DEFAULT 'Rs_Crores',

    UNIQUE (symbol, annual_end)
);

-- Corporate actions
CREATE TABLE corporate_actions (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol             TEXT NOT NULL REFERENCES stocks(symbol),
    action_date        DATE NOT NULL,
    action_type        TEXT NOT NULL,
    value              REAL,
    notes              TEXT,
    UNIQUE (symbol, action_date, action_type)
);

-- Technical indicators
CREATE TABLE technical_indicators (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol             TEXT NOT NULL REFERENCES stocks(symbol),
    date               DATE NOT NULL,
    close              REAL,
    rsi_14             REAL,
    macd               REAL,
    macd_signal        REAL,
    macd_hist          REAL,
    sma_50             REAL,
    sma_200            REAL,
    ema_21             REAL,
    bb_mid             REAL,
    bb_upper           REAL,
    bb_lower           REAL,
    atr_14             REAL,
    adx_14             REAL,
    vwap_14            REAL,
    obv                REAL,
    supertrend          REAL,
    supertrend_dir     INTEGER,
    UNIQUE (symbol, date)
);

-- Market indices

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CREATE TABLE market_indices (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    snapshot_date  DATE NOT NULL,
    index_name     TEXT NOT NULL,
    last_price     REAL, change_pct REAL, direction TEXT,
    UNIQUE (snapshot_date, index_name)
);

-- Forex & commodities
CREATE TABLE forex_commodities (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    snapshot_date  DATE NOT NULL,
    instrument     TEXT NOT NULL,
    last_price     REAL, change_pct REAL,
    UNIQUE (snapshot_date, instrument)
);

-- RBI rates
CREATE TABLE rbi_rates (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    effective_date DATE NOT NULL,
    repo_rate     REAL, reverse_repo REAL, sdf_rate REAL,
    msf_rate      REAL, bank_rate REAL, crr REAL, slr REAL,
    is_cached     INTEGER DEFAULT 0, source TEXT,
    UNIQUE (effective_date)
);

-- Macro indicators
CREATE TABLE macro_indicators (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    snapshot_date  DATE NOT NULL,
    indicator_name TEXT NOT NULL,
    source TEXT, value REAL, unit TEXT, year INTEGER, notes TEXT,
    UNIQUE (snapshot_date, indicator_name, year)
);

-- Ownership
CREATE TABLE ownership (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol         TEXT NOT NULL REFERENCES stocks(symbol),
    snapshot_date  DATE NOT NULL,
    promoter_pct   REAL,
    fii_fpi_pct   REAL,
    dii_pct       REAL,
    public_retail_pct REAL,
    num_shareholders INTEGER,
    insiders_pct   REAL,
    institutions_pct REAL,
    institutions_float_pct REAL,
    institutions_count INTEGER,
    total_institutional_pct REAL,
    fii_net_buy_cr REAL,
    dii_net_buy_cr REAL,
    fii_dii_flow_date TEXT,
    source         TEXT,
    UNIQUE (symbol, snapshot_date)
);

-- Ownership history
CREATE TABLE ownership_history (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol         TEXT NOT NULL REFERENCES stocks(symbol),
    period_end     DATE NOT NULL,
    promoter_pct   REAL NOT NULL,
    fii_pct       REAL,
    dii_pct       REAL,
    public_pct     REAL,
    total_institutional_pct REAL,
    num_shareholders INTEGER,
    source         TEXT DEFAULT 'Screener.in',
    UNIQUE (symbol, period_end)
);

-- Earnings history
CREATE TABLE earnings_history (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol         TEXT NOT NULL REFERENCES stocks(symbol),
    quarter_end    DATE NOT NULL,
    eps_actual     REAL, eps_estimate REAL,

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        eps_difference REAL, surprise_pct REAL,
        UNIQUE (symbol, quarter_end)
    );

-- Earnings estimates
CREATE TABLE earnings_estimates (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL REFERENCES stocks(symbol),
    snapshot_date DATE NOT NULL,
    period_code TEXT NOT NULL,
    avg_eps REAL, low_eps REAL, high_eps REAL,
    year_ago_eps REAL, analyst_count INTEGER, growth_pct REAL,
    UNIQUE (symbol, snapshot_date, period_code)
);

-- EPS trend
CREATE TABLE eps_trend (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL REFERENCES stocks(symbol),
    snapshot_date DATE NOT NULL,
    period_code TEXT NOT NULL,
    current_est REAL, seven_days_ago REAL, thirty_days_ago REAL,
    sixty_days_ago REAL, ninety_days_ago REAL,
    UNIQUE (symbol, snapshot_date, period_code)
);

-- EPS revisions
CREATE TABLE eps_revisions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL REFERENCES stocks(symbol),
    snapshot_date DATE NOT NULL,
    period_code TEXT NOT NULL,
    up_last_7d INTEGER, up_last_30d INTEGER,
    down_last_30d INTEGER, down_last_7d INTEGER,
    UNIQUE (symbol, snapshot_date, period_code)
);

-- Data quality log
CREATE TABLE data_quality_log (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    run_timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    symbol TEXT NOT NULL,
    table_name TEXT NOT NULL,
    rows_inserted INTEGER DEFAULT 0,
    rows_null_heavy INTEGER DEFAULT 0,
    avg_completeness REAL,
    critical_nulls_json TEXT,
    source TEXT,
    notes TEXT
);

-- Run log
CREATE TABLE run_log (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL,
    run_timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    script_version TEXT,
    modules_ok TEXT,
    modules_warn TEXT,
    notes TEXT
);

-- Growth metrics
CREATE TABLE "growth_metrics"(
    id INT,
    symbol TEXT,
    as_of_date NUM,
    revenue_cagr_3y REAL,
    net_profit_cagr_3y REAL,
    ebitda_cagr_3y REAL,
    eps_cagr_3y REAL,
    fcf_cagr_3y REAL,
    sales_cagr_10y REAL,
    sales_cagr_5y REAL,
    sales_cagr_3y REAL,
    sales_ttm REAL,
    profit_cagr_10y REAL,
    profit_cagr_5y REAL,
    profit_cagr_3y REAL,

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    profit_ttm REAL,
    stock_cagr_10y REAL,
    stock_cagr_5y REAL,
    stock_cagr_3y REAL,
    stock_ttm REAL,
    growth_available INT,
    completeness_pct REAL,
    roe_10y REAL,
    roe_5y REAL,
    roe_3y REAL,
    roe_last REAL
);

-- Balance sheet
CREATE TABLE balance_sheet (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL,
    period_end DATE NOT NULL,
    period_type TEXT NOT NULL DEFAULT 'annual',
    equity_capital REAL,
    reserves REAL,
    total_equity REAL,
    borrowings REAL,
    lt_borrowings REAL,
    st_borrowings REAL,
    lease_liabilities REAL,
    preference_capital REAL,
    other_borrowings REAL,
    other_liabilities REAL,
    minority_interest REAL,
    trade_payables REAL,
    advance_from_customers REAL,
    other_liability_items REAL,
    total_liabilities REAL,
    fixed_assets REAL,
    cwip REAL,
    investments REAL,
    other_assets REAL,
    inventories REAL,
    trade_receivables REAL,
    cash_equivalents REAL,
    loans_advances REAL,
    other_asset_items REAL,
    total_assets REAL,
    net_debt REAL,
    data_source TEXT DEFAULT 'screener',
    completeness_pct REAL,
    missing_fields_json TEXT,
    UNIQUE(symbol, period_end, period_type)
);

-- Cash flow
CREATE TABLE cash_flow (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL REFERENCES stocks(symbol),
    period_end DATE NOT NULL,
    period_type TEXT NOT NULL DEFAULT 'annual',
    cfo REAL,
    cfi REAL,
    cff REAL,
    capex REAL,
    free_cash_flow REAL,
    net_cash_flow REAL,
    raw_details_json TEXT,
    data_source TEXT DEFAULT 'screener',
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    completeness_pct REAL,
    missing_fields_json TEXT,
    UNIQUE(symbol, period_end, period_type)
);

-- Profit and loss (legacy/alternative table)
CREATE TABLE profit_and_loss (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL,
    period_end DATE NOT NULL,
    period_type TEXT NOT NULL DEFAULT 'annual',
    sales REAL,
    expenses REAL,

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    operating_profit REAL,
    opm_pct TEXT,
    other_income REAL,
    interest REAL,
    depreciation REAL,
    profit_before_tax REAL,
    tax_pct TEXT,
    net_profit REAL,
    eps REAL,
    dividend_payout_pct REAL,
    is_interpolated INTEGER DEFAULT 0,
    data_source TEXT DEFAULT 'screener',
    completeness_pct REAL,
    missing_fields_json TEXT
);
-- System table (optional, created automatically by SQLite)
CREATE TABLE sqlite_sequence(name,seq);

-- Indexes
CREATE INDEX idx_price_sym_date ON price_daily(symbol, date DESC);
CREATE INDEX idx_price_intra_sym ON price_intraday(symbol, ts DESC);
CREATE INDEX idx_qr_sym ON quarterly_results(symbol, period_end DESC);
CREATE INDEX idx_ar_sym ON annual_results(symbol, period_end DESC);
CREATE INDEX idx_qcd_sym ON "annual_cashflow_derived"(symbol, annual_end DESC);
CREATE INDEX idx_ca_sym ON corporate_actions(symbol, action_date DESC);
CREATE INDEX idx_ti_sym ON technical_indicators(symbol, date DESC);
CREATE INDEX idx_own_hist ON ownership_history(symbol, period_end DESC);
CREATE INDEX idx_dql_sym ON data_quality_log(symbol, run_timestamp DESC);
CREATE INDEX idx_bs_sym ON balance_sheet(symbol, period_type, period_end DESC);
CREATE INDEX idx_cf_sym_date ON cash_flow(symbol, period_end DESC);
CREATE INDEX idx_cf_sym ON cash_flow(symbol, period_type, period_end DESC);
-- Unique indexes
CREATE UNIQUE INDEX growth_metrics_symbol_date ON growth_metrics(symbol, as_of_date);
CREATE UNIQUE INDEX idx_pnl_unique ON profit_and_loss(symbol, period_end, period_type);

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